ALICE L WALTON SCHOOL OF MEDICINE

Platelet Assays

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Adapted by: Chelsey Deel
Editor: Jordan Finn
Adapted from: Platelet Assays

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Learning Objectives (5) Versions

After completing this brick, you will be able to:

1. Explain the purpose of platelet assays.
2. Describe the clinical utility of the platelet count.
3. Describe the clinical utility of assessing platelet morphology.
4. Explain the platelet function assay, its uses, and its limitations.
5. Explain how [illegible] describe the conditions for which they are used for diagnosis.

CASE CONNECTION

BT is a 27-year-old female referred to you by her oral surgeon. After wisdom teeth removal, BT had continuous oozing at the extraction sites. “I’m not sure what the problem is. I checked her prothrombin time/international normalized ratio, partial thromboplastin time, and platelet count, but that’s as far as I can go,” the oral surgeon says. You review the lab report and note that these test results are all normal, including a platelet count within the normal range.

What test will you perform to diagnose BT’s condition? Consider your answer as you read, and we’ll revisit BT at the end of the brick.

GO TO CONCLUSION ↓

What Are Platelet Assays (LO #1)?

Platelet assays are tests used to study whether there is a defect in the number or function of platelets. They usually are performed when a patient has abnormal bleeding.

Why do we test the platelets? Remember, if you want to make a blood clot, you need to do two things: make a platelet plug and seal that plug up with fibrin. Platelet plug formation is called primary hemostasis, and fibrin formation, which occurs by means of the coagulation cascade, is called secondary hemostasis. Defects i[illegible]asis can lead to poor clot formation and excessive bleeding.

So, if a patient presents with unexplained bleeding—or if we just want to assess how well a patient can clot (before performing surgery, for example)—the two main things to evaluate are the platelets and the coagulation cascade. In this brick, we focus specifically on platelet assays, tests that evaluate platelet counts, morphology, and function.

Why is assessing the coagulation cascade important in a patient undergoing surgery?

What Is the Platelet Count (LO #2)?

A platelet count is a lab test used to determine the number of platelets in the blood. The test may be used to evaluate a patient with a coagulation disorder (either too much bleeding or too much clotting), but it also is used during the evaluation of any type of hematology disorder (e.g., leukemias, anemias), because a deficiency [illegible]mpanied by a deficiency of another.

A normal platelet count is 150,000 to 400,000 platelets/mm^3 of blood. The clinical implications of certain platelet counts are illustrated in Table 1.

Table 1. Platelet Counts

Platelet Count	Clinical Implication
>400,000/mm^3	**Thrombocytosis.** Patients with extremely high numbers of platelets may experience ischemic events, because the blood gets “sludgy” and flows slowly when there are so many platelets present.
<150,000/mm^3	**Thrombocytopenia.** Usually no significant bleeding with mild thrombocytopenia.
<50,000/mm^3	May start to see some abnormal bleeding
20,000-50,000/mm^3	May lead to increased bleeding after injuries
<10,000/mm^3	Risk of spontaneous bleeding

Hematology analyzer technology has been advancing, with sophisticated devices able to automatically provide reliable platelet counts. Nevertheless, platelet counts can be falsely elevated or decreased when other small cells are misread as platelets or when platelet clumps are present. In these instances, platelet counts must be performed manually by examining a blood smear under the microscope.

What is the difference between thrombocytosis and thrombocytopenia?

How Do We Assess Platelet Morphology (LO #3)?

In contrast to the platelet count, assessment of platelet morphology requires manual evaluation of a blood smear under a microscope. This can reveal subtle morphologic abnormalities in the platelets that autoanalyzer devices cannot detect. Such an analysis includes looking at the physiological and pathological features of the platelets, including size, shape, and granulation.

Clinically, morphology analysis is used less frequently than is the total platelet count because most diseases—even those that directly involve platelets—have normal-appearing platelets (Figure 1, red arrows).

CLINICAL CORRELATION

However, in certain [illegible]part of the diagnosis. Hereditary platelet disorders, for example, may have giant platelets (Figure 1, blue arrow) or platelets that lack granules and appear gray.

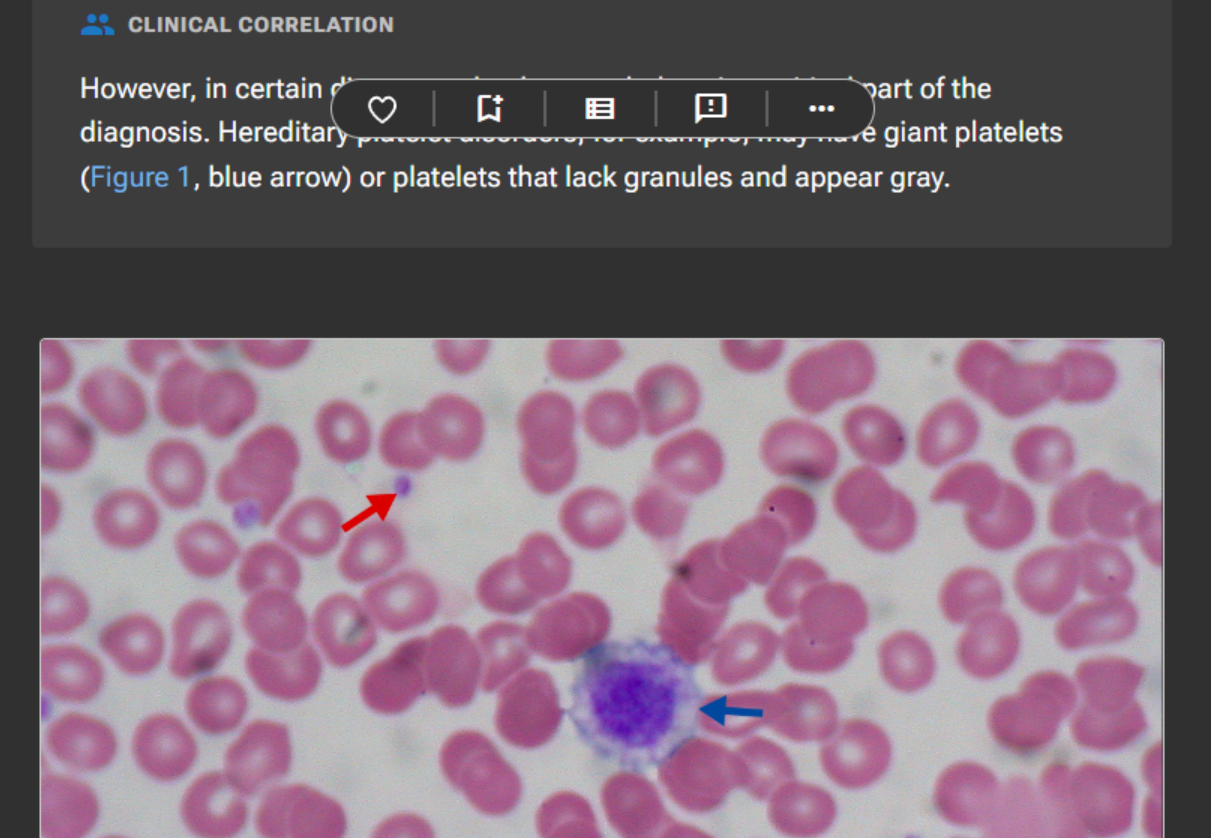

Figure 1

What are abnormal platelet morphologies seen on the blood smear?

How Is Platelet Function Evaluated (LO #4)?

We discussed how the platelet count and morphology can be useful clinically. However, these assays don’t tell us anything about how well the platelets actually work. Fortunately, there are several tests that can be used to measure platelet function.

Platelet Function Assay

The preferred test used to determine platelet function is the platelet function assay (PFA-100). In this test, blood is fed into a machine with specific chemicals that cause platelets to form a plug over a standardized aperture. The test measures how well the platelet plug forms (primary hemostasis) and can detect both abnormal platelet adhesion and aggregation.

Bleeding Time

The PFA is now replacing the older test of platelet function called the bleeding time. Bleeding time is performed by using a small lancet to pierce the skin of the forearm and bloti[illegible]n to form; the provider records the time it takes for the cut to stop bleeding. The time it takes from beginning to end represents the time it takes a person to perform primary hemostasis (the platelet plug). The reference range for bleeding time is usually 2 to 9 minutes.

The bleeding time has limits, which is why it is being replaced by the PFA-100. It is difficult to standardize the test because of variables such as the size of the lancet, the skin depth and durability, and the technician’s blotting technique.

Limitations

Neither of these tests can explain why the platelets are not performing primary hemostasis adequately. For example, patients can have a deficiency of platelet receptors, as in Bernard-Soulier syndrome, or a deficiency of von Willebrand factor (vWF), to name just a couple reasons. In addition, neither test reflects the activity of fibrin and the formation of a mature clot.

Why is it that neither the PFA nor the bleeding time test indicates why the platelets are not performing primary hemostasis adequately?

What Are Tests to Determine the Cause of Platelet Dysfunction (LO #5)?

So now we know from the above-described functional studies that the platelets are functioning poorly, but we may not know why. Determining why is done using platelet aggregation studies showing platelet response to specific factors, as well as by using more specialized tests, including measurement of vWF directly.

Platelet Aggregation Studies

These tests complement the functional studies described above and show how well the patient’s platelets aggregate in response to different aggregating agents.

The principle behind the test is pretty simple: add an aggregating agent to a tube containing platelet-rich plasma. As the platelets clump together, the plasma will become less cloudy, or turbid (Figure 2).

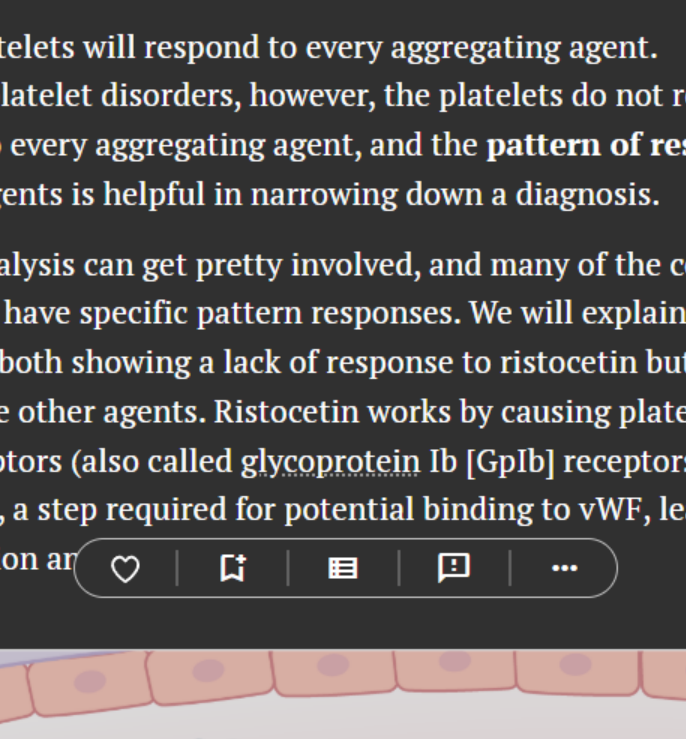

Figure 2

A decrease in turbidity means that the platelets are aggregating properly in response to that particular agent.

The test is repeated with several **different aggregating agents**, such as ristocetin, collagen, adenosine diphosphate (ADP), and epinephrine:

- Normal platelets will respond to every aggregating agent.
- In certain platelet disorders, however, the platelets do not respond normally to every aggregating agent, and the **pattern of response** to the different agents is helpful in narrowing down a diagnosis.

This pattern analysis can get pretty involved, and many of the congenital platelet defects have specific pattern responses. We will explain a couple examples here, both showing a lack of response to ristocetin but normal responses to the other agents. Ristocetin works by causing platelets to move more vWF receptors (also called glycoprotein Ib [GpIb] receptors) to the platelet surface, a step required for potential binding to vWF, leading to platelet activation ar[illegible]

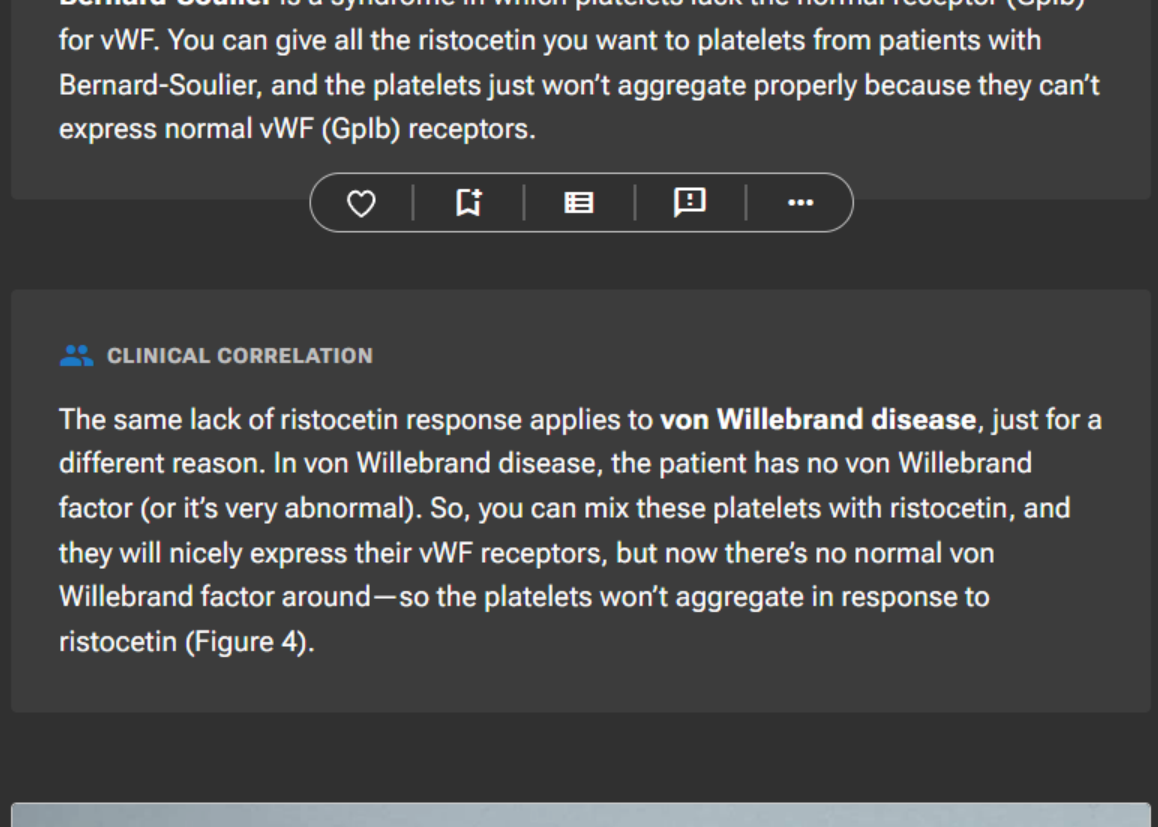

Figure 3

CLINICAL CORRELATION

Bernard-Soulier is a syndrome in which platelets lack the normal receptor (GpIb) for vWF. You can give all the ristocetin you want to platelets from patients with Bernard-Soulier, and the platelets just won’t aggregate properly because they can’t express normal vWF (GpIb) receptors.

CLINICAL CORRELATION

The same lack of ristocetin response applies to **von Willebrand disease**, just for a different reason. In von Willebrand disease, the patient has no von Willebrand factor (or it’s very abnormal). So, you can mix these platelets with ristocetin, and they will nicely express their vWF receptors, but now there’s no normal von Willebrand factor around—so the platelets won’t aggregate in response to ristocetin (Figure 4).

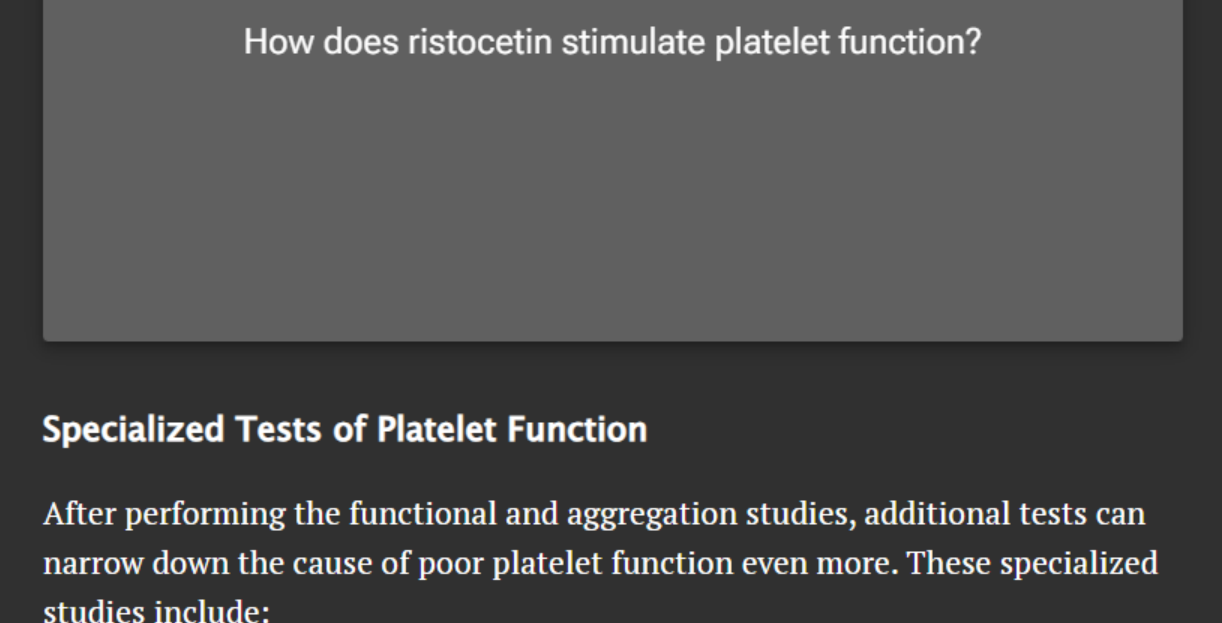

Figure 4. Platelet aggregation studies. This patient shows normal aggregation to all agonists apart from ristocetin. This may represent Von Willebrand disease or Bernard-Soulier syndrome.

How does ristocetin stimulate platelet function?

Specialized Tests of Platelet Function

After performing the functional and aggregation studies, additional tests can narrow down the cause of poor platelet function even more. These specialized studies include:

- Von Willebrand factor assays (“von Willebrand antigen”); the level will be low in von Willebrand disease.
- Flow cytometry studies using whole blood. These can assess platelet membrane markers such as the glycoprotein (Gp) complexes shown in Figure 3, which may be decreased in some disorders.

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to BT, what test do you perform to diagnose BT’s condition?

“So, I have enough platelets, but they’re just not working well?” BT asks you after you discuss why you think she is having abnormal bleeding. “That’s about right,” you reply. “I want to do some special tests that will help me determine the particular problem.” You decide to do a PFA-100 test, which shows decreased platelet function. As a follow-up, a platelet aggregation shows normal response to collagen, adenosine diphosphate, and epinephrine. However, in the ristocetin stimulation test, the platelets do not aggregate. Additional testing shows a low level of von Willebrand factor, indicating that BT has von Willebrand disease. You explain this to BT and discuss treatment with her, especially with regard to risks for any surgery she might have in the future.

Summary

- Platelet assays are tests that measure platelet counts, analyze platelet morphology, and evaluate how well platelets aggregate and promote clotting.
- These assays can be especially useful in assessing patient clotting function preoperatively.
- Measuring platelet count is an initial step in determining the cause of abnormal bleeding.
- It is helpful to monitor platelet count in drug therapies to avoid adverse reactions or monitor progress in certain cancer treatments.
- The platelet count does not indicate platelet function; a patient can have a bleeding disorder and still have a normal platelet count.
- Platelet morphology is evaluated using a blood smear.
- Most diseases have normal-appearing platelets, but in some diseases, such as hereditary plat[illegible]phology is key to diagnosis.
- The platelet function assay PFA-100 test is the preferred test to measure platelet adhesion and aggregation.
- The PFA-100 has supplanted the in vivo bleeding time as the first measurement of platelet function.
- Platelet aggregation assays reveal platelet response to biochemical agonists and help differentiate bleeding disorders.
- Specialized tests, such as von Willebrand factor (vWF) assays and flow cytometry, can determine the specific cause of reduced platelet function.

Review Questions

1. A 42-year-old male who had a bone marrow transplant 10 days ago has thrombocytopenia. His platelet count on Monday was 120,000/mm^3, and his count on Wednesday was 100,000/mm^3. If his platelet count continues to drop, which of the following platelet counts is most associated with the onset of spontaneous bleeding?

A. ● 10,000/mm^3
B. ○ 40,000/mm^3
C. ○ 60,000/ mm^3
D. ○ 80,000/mm^3

Hide Explanation

The correct answer is 10,000/mm^3 (A). Severe or spontaneous bleeding typically does not happen until the platelet count dips below 10,000/mm^3. Patients with counts of 40,000/mm^3 (B) may have increased bleeding with trauma or injury. Counts of 60,000/mm^3 (C) and 80,000/mm^3 (D) are not usually associated with severe bleeding.

2. A sample of platelet-rich plasma is sent to the lab for aggregation studies. The sample is exposed to a variety of agonists and shows normal platelet aggregation under all conditions. Which of the following conclusions can be drawn from these results?

A. ○ The patient has abnormal platelet function.
B. ○ The patient has a sufficient quantity of platelets to perform primary hemostasis.
C. ○ The patient has either Bernard-Soulier syndrome or von Willebrand factor deficiency.
D. ● The patient has normal-functioning platelets.
E. ○ The patient will be able to perform secondary hemostasis.

Hide Explanation

The correct answer is the patient has normal-functioning platelets (D). The sample shows the ability to a[illegible]hosphate, ristocetin, collagen, etc.) of any [illegible]dered to be a normal function in aggregation studies. Patients with abnormal platelet function would have diminished platelet aggregation in response to one or more of the aggregating agents (A). Aggregation studies do not reveal the quantity of platelets (B). Although abnormal platelet aggregation studies would help differentiate between Bernard-Soulier syndrome and von Willebrand factor deficiency, the normal results in this case preclude either diagnosis (C). Platelet aggregation studies do not assess the formation of fibrin (secondary hemostasis) (E).

3. Which of the following is the best measure of platelet function?

A. ○ Mean platelet volume
B. ○ Platelet count
C. ○ Platelet morphology
D. ● PFA-100

Hide Explanation

The correct answer is PFA-100 (D). The test is an in vitro assay of platelet function, estimating both platelet aggregation and adhesion (part of primary hemostasis). It is replacing the traditional bleeding time as the main measure of platelet function. Mean platelet volumes (A) and platelet counts (B) are measured on a flow cytometer, and they reveal the number and size of the platelets (which is not necessarily reflective of the actual functioning of the platelets). Platelet morphology (C), as seen on a blo[illegible]iagnostic information—but in most cases, it does not correlate with the function of the platelets.

4. What of the following platelet morphologies is seen in most diseases?

A. ○ Abnormally large platelets
B. ○ Abnormally small platelets
C. ● Normal platelet morphology
D. ○ Platelets lacking granules

Hide Explanation

The correct answer is normal platelet morphology (C). In most diseases, even those involving platelets, the morphology of the platelets is normal, limiting the utility of peripheral smear evaluation. Platelets that are abnormally large (A), abnormally small (B), or lack granules (D) are suggestive of more specific diseases.

How would you rate this Brick?

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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Explain the purpose of platelet assays.~~
02 ☑ ~~Describe the clinical utility of the platelet count.~~

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04 ☑ ~~Explain the platelet function assay, its uses, and its limitations.~~
05 ☑ ~~Explain how platelet aggregation studies work and describe the conditions for which they are used for diagnosis.~~

Objective Confidence: 5/5